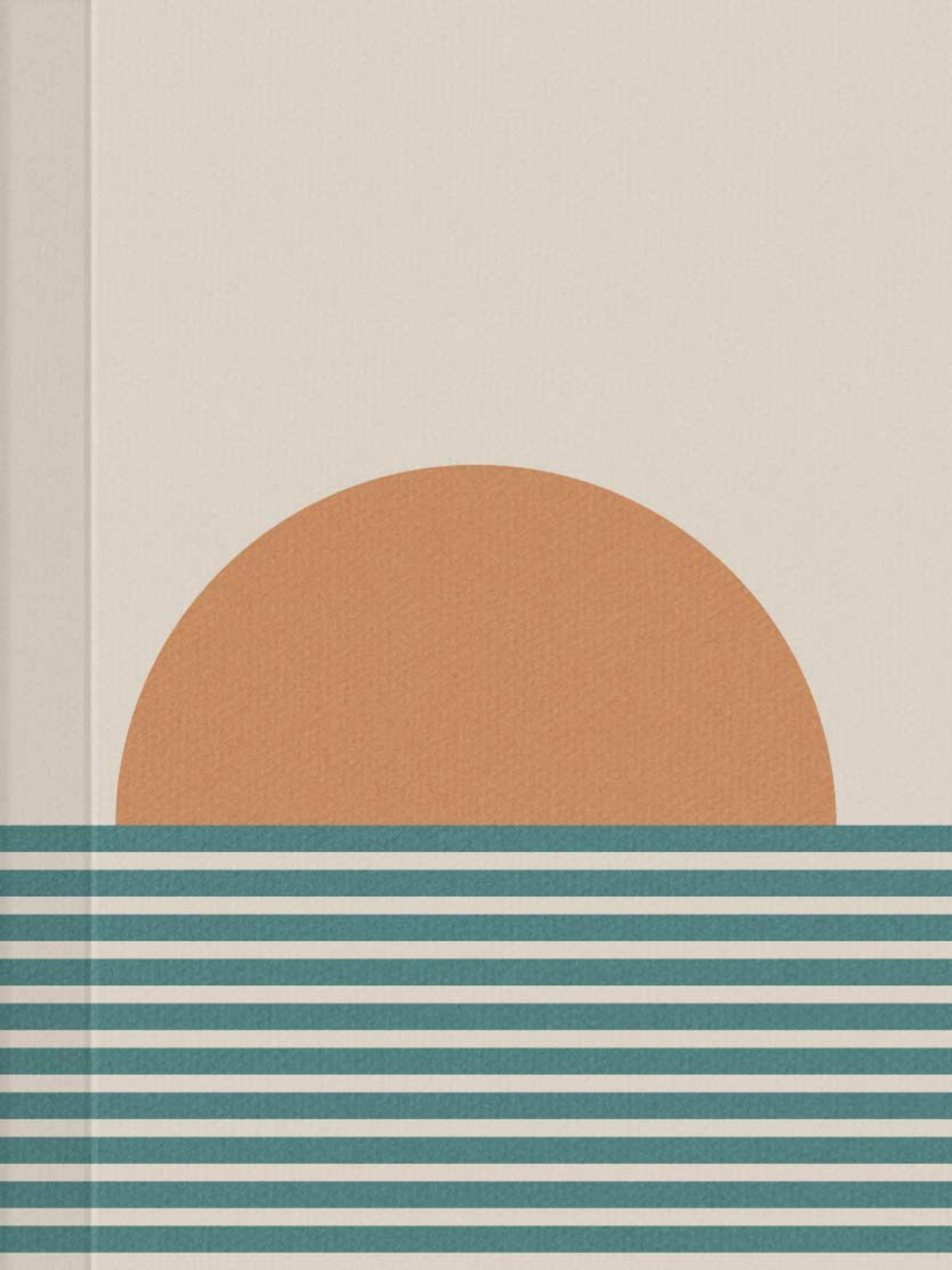




23/02 Claude Shannon

Probabilidade $\uparrow$ Surpresa $\downarrow$ Informação de Shannon
(Surpresa)Probabilidade $\downarrow$ Surpresa $\uparrow$

$$S(x) = \frac{1}{P(x)}$$

$$\begin{aligned} \checkmark S(x_1, x_2, x_3) &= \frac{1}{P(x_1, x_2, x_3)} = \frac{1}{P(x_1) \cdot P(x_2) \cdot P(x_3)} \\ &= \frac{1}{P(x_1)} \times \frac{1}{P(x_2)} \times \frac{1}{P(x_3)} \\ &\quad \downarrow \quad \quad \downarrow \quad \quad \downarrow \\ &\quad S(x_1) + S(x_2) + S(x_3) \end{aligned}$$

Entropia (Medida de incerteza)

$$\begin{aligned} \checkmark H(x) &= E[S(x)] \\ &= \sum_x P(x) \cdot \log \frac{1}{P(x)} \end{aligned}$$

Informação Mútua (Medida de dependência)

$$\checkmark I(x; Y) = \sum_{x,y} P(x,y) \cdot \log \frac{P(x,y)}{P(x) \cdot P(y)}$$

Divergência de Kullback-Leibler

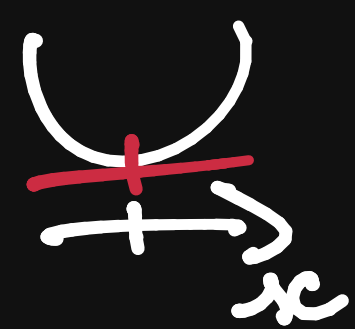
$$D(p \parallel q) = \sum_x p(x) \cdot \log \frac{p(x)}{q(x)}$$

Se $p(x) = q(x)$ então $D(p \parallel q) = 0$

		Sol	
		s	N
Chuva	s	0.1	0.2
	N	0.5	0.2
		0.6	0.4

$$\begin{aligned} I(x; Y) &= 0.1 \times \log \left(\frac{0.1}{0.6 \times 0.3} \right) + 0.2 \times \log \left(\frac{0.2}{0.4 \times 0.3} \right) + 0.5 \times \log \left(\frac{0.5}{0.6 \times 0.7} \right) + \\ &\quad + 0.2 \times \log \left(\frac{0.2}{0.4 \times 0.7} \right) \approx 0.091 \text{ Sh (Shannon)} \end{aligned}$$

09/03 $f(x) = x^2 - 4x + 3$
 Mínimo da função



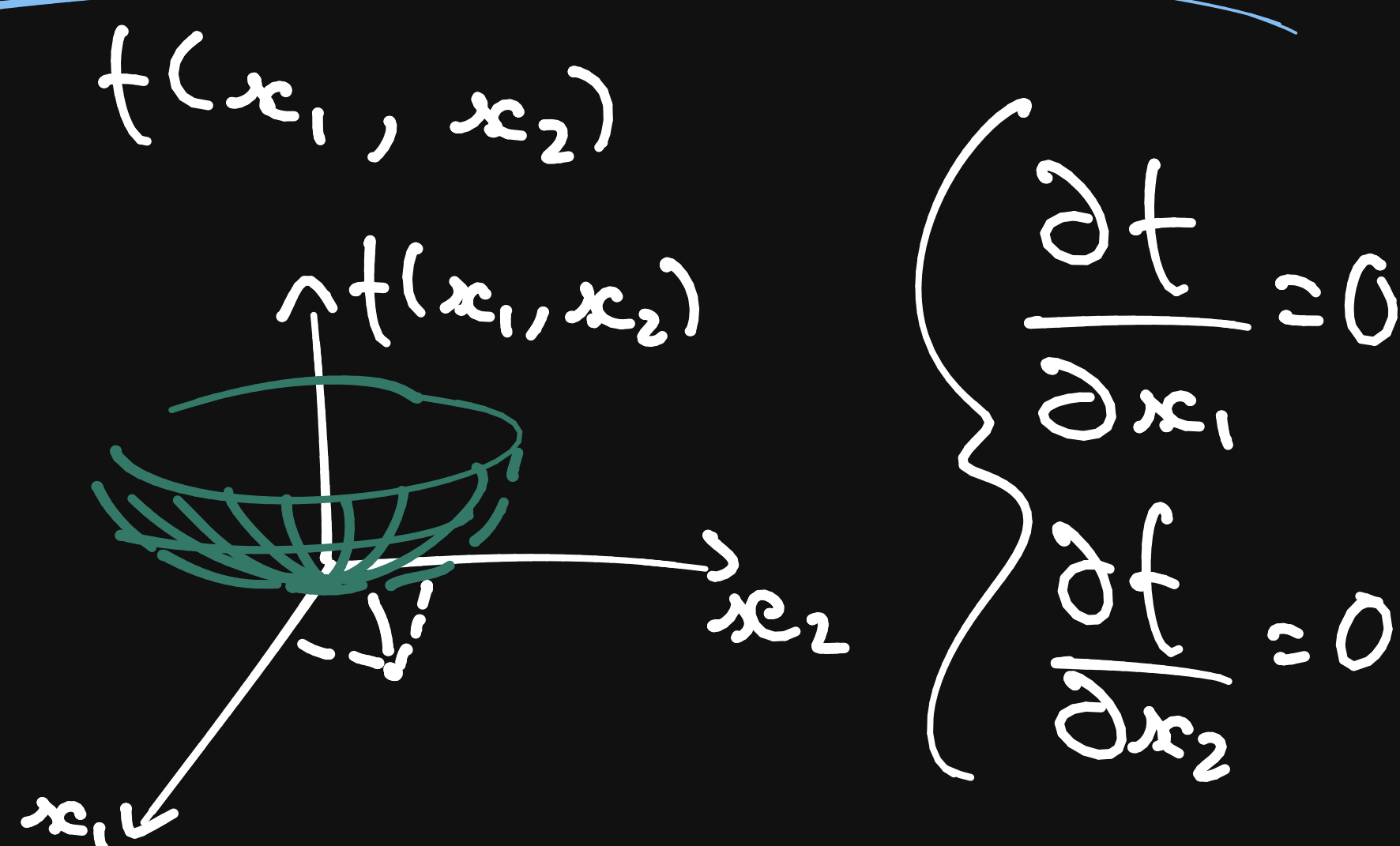
$$\frac{df}{dx} = 2x - 4$$

$$\Delta x = -\eta \frac{df}{dx}$$

$$\Rightarrow x_k - x_{k-1} = -\eta \frac{df}{dx}$$

$$\Rightarrow x_k = x_{k-1} - \eta \frac{df}{dx}$$

↑
tamanho
do passo



$$f(x_1, x_2) = x_1^2 + 3x_2^2 + 2x_1x_2 + 1$$

$$\begin{cases} \frac{\partial f}{\partial x_1} = 2x_1 + 2x_2 = 0 \\ \frac{\partial f}{\partial x_2} = 6x_2 + 2x_1 = 0 \end{cases}$$

Solução: Mínimo em $(x_1, x_2) = (0, 0)$

Inicial = $(-5, 7)$

$$\eta = 0.1$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -5 \\ 7 \end{bmatrix} - 0.1 \times \begin{bmatrix} 4 \\ 32 \end{bmatrix} = \begin{bmatrix} -5.4 \\ 3.8 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -5.4 \\ 3.8 \end{bmatrix} - 0.1 \times \begin{bmatrix} -3.2 \\ 12 \end{bmatrix} = \dots$$

Exemplo:

$$\eta = 0.1$$

$$x_0 = 10$$

$$x_1 = x_0 - \eta \frac{df}{dx}(x_0) =$$

$$= 10 - 0.1 \times 16 = \underline{\underline{8.4}}$$

$$x_2 = 8.4 - 0.1 \times 12.8 = \underline{\underline{7.12}}$$

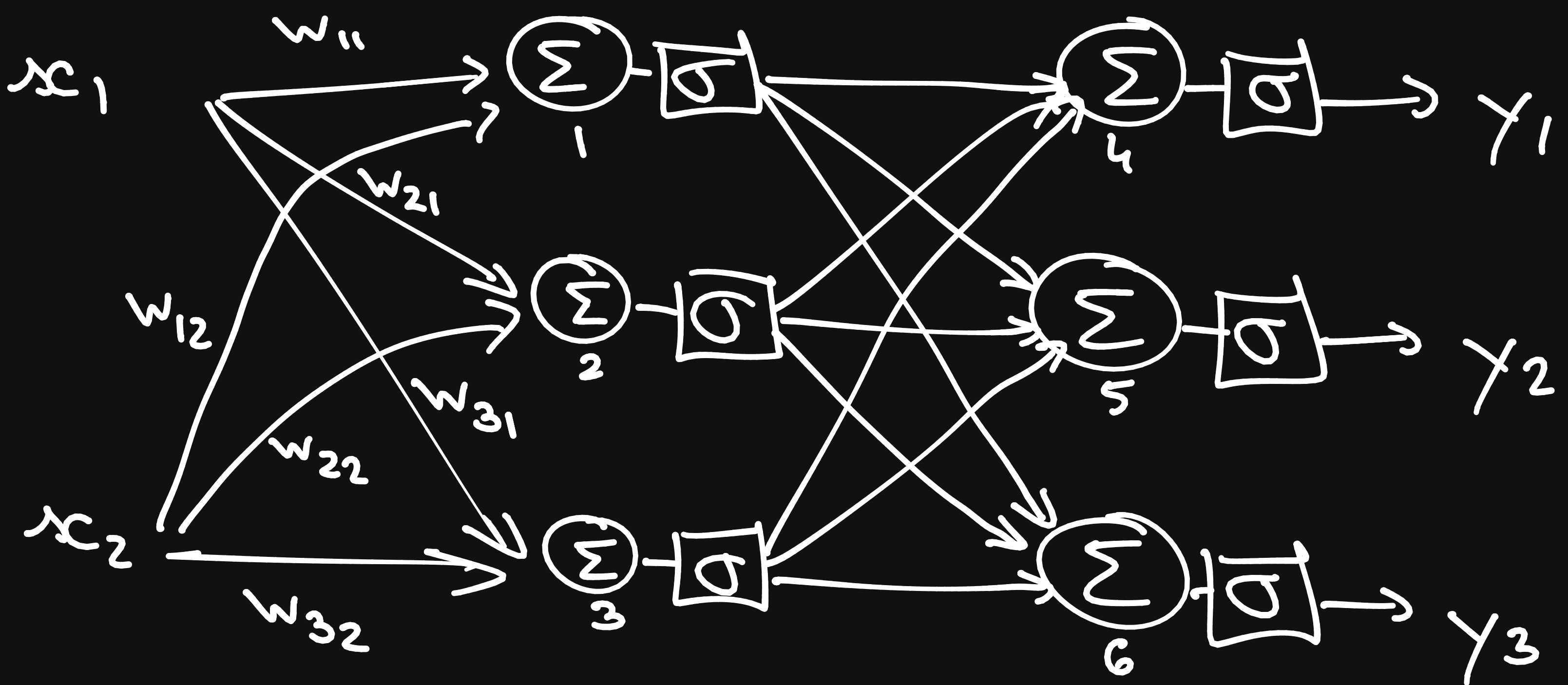
$$x_3 = 7.12 - 0.1 \times 10.24 = \underline{\underline{6.10}}$$

η passo

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \leftarrow \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \eta \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \end{bmatrix}$$

Método
do Gradiente

16/04



treino $\{(x, y^*)\}$

$$E = \frac{1}{2} \sum (y_n^* - y_n)^2$$

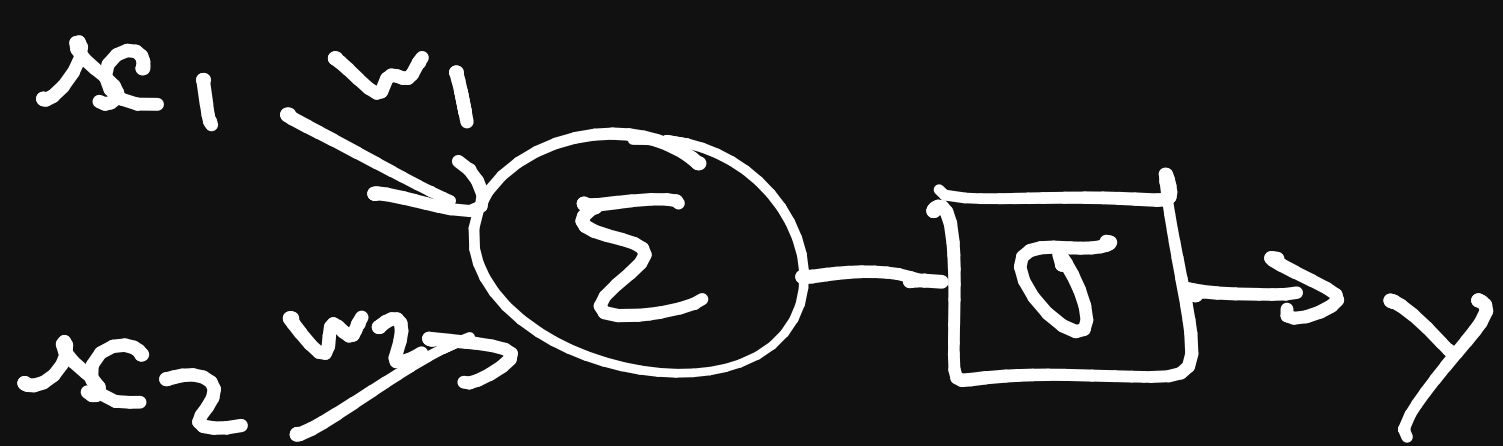
$$\begin{cases} \frac{\partial E}{\partial w_{11}} = 0 \\ \frac{\partial E}{\partial w_{12}} \\ \dots \end{cases}$$

$$\frac{\partial E}{\partial w_{ij}} = 0 \text{ Não se consegue}$$

$$\Delta w_{ji} = -\eta \frac{\partial E}{\partial w_{ji}}$$

Método do Gradiente

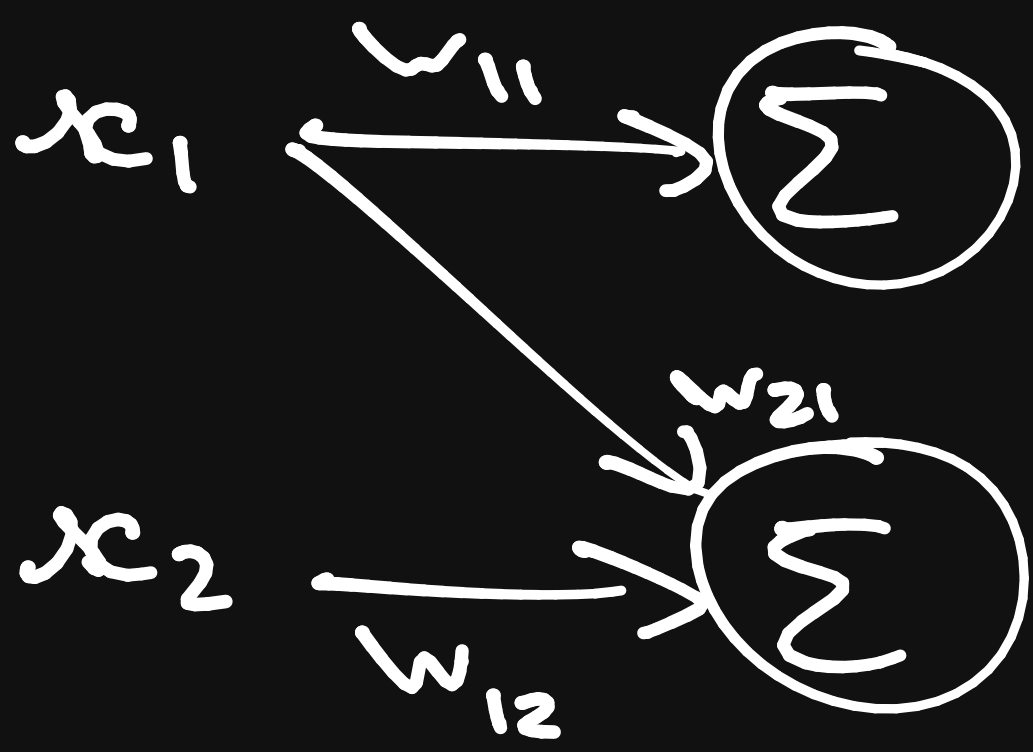
1 Neurônio



$$\frac{\partial E}{\partial w_1} = \frac{\partial E}{\partial w_1} \left[\frac{1}{2} \sum_n (y_n^* - y_n)^2 \right] = \frac{1}{2} \frac{\partial}{\partial w_1} \sum_n (y_n^* - y_n)^2 =$$

$$= \frac{1}{2} \sum_n \frac{\partial}{\partial w_1} (y_n^* - y_n)^2 = \cancel{\frac{1}{2}} \sum_n \cancel{2} (y_n^* - y_n) \frac{\partial}{\partial w_1} (y_n^* - y_n) =$$

$$= \sum_n (y_n^* - y_n) \frac{\partial}{\partial w_1} \sigma(x_1 w_1 + x_2 w_2) = - \sum_n (y_n^* - y_n) \sigma''(x_1 w_1 + x_2 w_2)$$



23/03 Função Ativação

$\sigma(x)$ Função logística/sigmoid

$$\sigma(x) = \frac{1}{1 + \exp(-x)}$$

	Regressão	Classificação
Ultimo Layer	Identidade $*$	
Função Erro $E(x)$?	$\ y - y^*\ ^2$	

$$E(w) = \frac{1}{2} \sum \|y(x, w) - y^*\|^2$$

$$N(y^* | y(x, w), \sigma^2)$$

com N observações independentes

$$P(y^* | x, w, \sigma^2) = \prod_{i=1}^N p(y_n^* | x_n, w, \sigma^2)$$

↙ -log

$$= - \sum_{n=1}^N \log p(y_n^* | x_n, w, \sigma^2)$$

$$= \frac{1}{2} \frac{\sum_n (y(x_n, w) - y_n^*)^2}{\sigma^2}$$

$$N(x | \eta, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2} \frac{(x - \eta)^2}{\sigma^2}}$$

$$y = \frac{1}{1 + e^{-x}} \quad \text{interpretar como uma possibilidade}$$

$$P(C_1 | x), \quad P(C_2 | x) = 1 - p(C_1 | x)$$

A distribuição condicional de y^* dados os inputs x tem a distribuição de Bernoulli:

$$P(y^* | x, w) = \underbrace{y(x, w)^{y^*}}_{P(C_1 | x)} \underbrace{(1 - y(x, w))^{1 - y^*}}_{P(C_2 | x)}$$

$y^* \text{ é } 0 \text{ ou } 1$

$$\prod_{n=1}^N P(y^* | x, w)$$

$\downarrow -\log$

$$y^* \log y(x, w) + (1 - y^*) \log (1 - y(x, w))$$

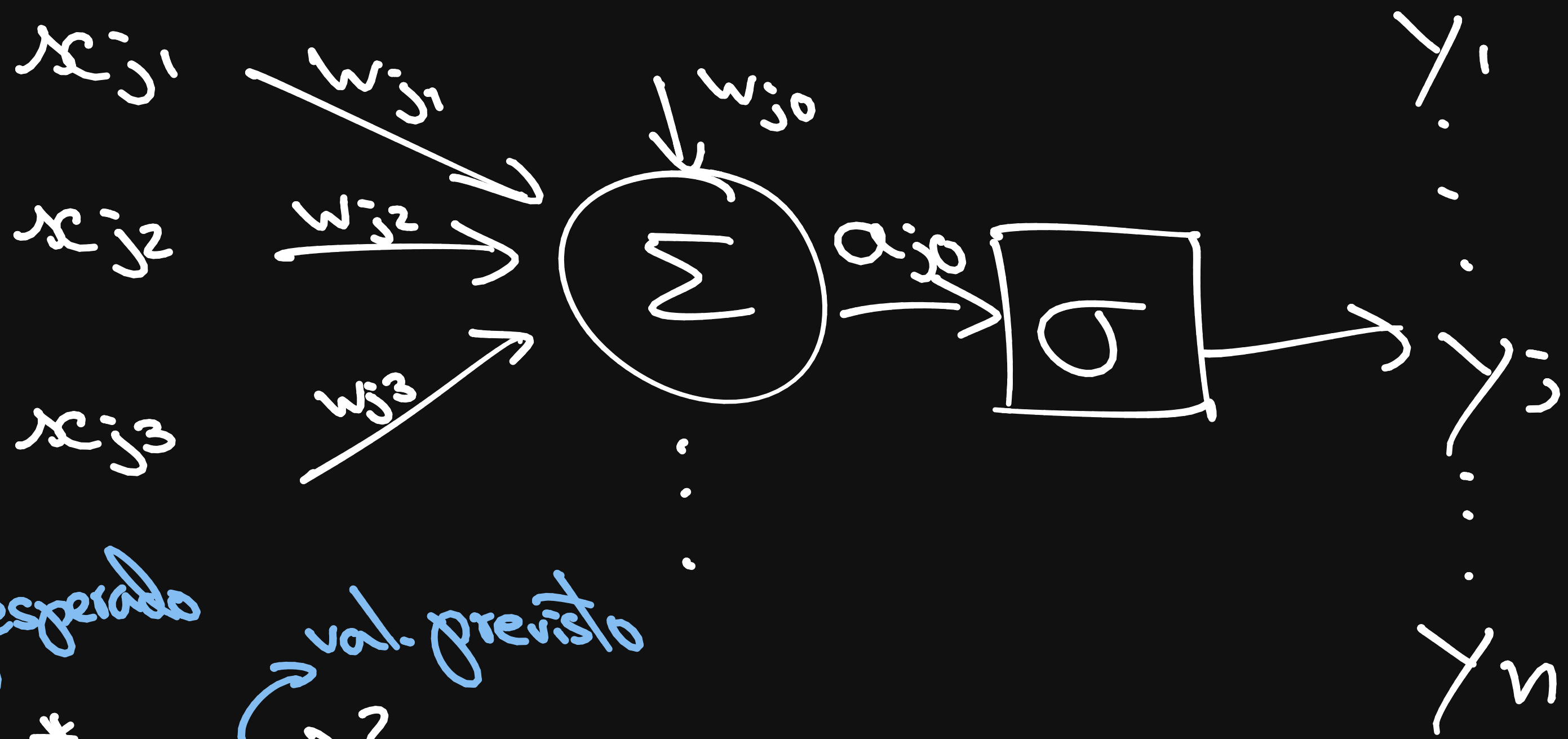
Minimizador Entropia Relativa

Entropia Relativa (Divergência Kullback

$$\begin{aligned} D(p \parallel q) &= \sum_x p(x) \log \frac{p(x)}{q(x)} = \\ &= \sum_x p(x) \log p(x) - \underbrace{\sum_x p(x) \log q(x)}_{-H(x)} \end{aligned}$$

13/04

Prática



$$E = \frac{1}{2} \sum_n (\overset{\text{val. esperado}}{y_n^*} - \overset{\text{val. previsto}}{y_n})^2$$

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

depende
de w_{ji}

Calcular

$$\frac{\partial E}{\partial w_{ji}}$$

Função ativação /

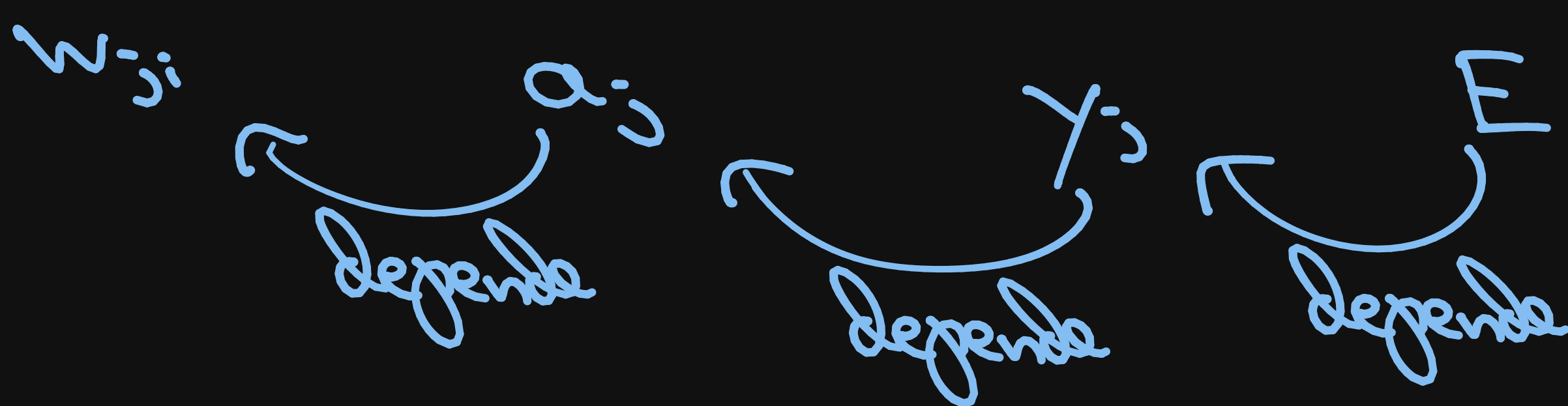
Saída do neurónio j

$$y_j = \sigma(a_j)$$

Soma ponderada /

Entrada do neurónio j

$$a_j = \sum_i w_{ji} x_{ji}$$



$$\frac{\partial E}{\partial w_{ji}} = \frac{\partial}{\partial w_{ji}} \left[\frac{1}{2} \sum_n (y_n^* - y_n)^2 \right] =$$

$$= \frac{1}{2} \frac{\partial}{\partial w_{ji}} \sum_n (y_n^* - y_n)^2 =$$

$$= \frac{1}{2} \sum_n \frac{\partial}{\partial w_{ji}} (y_n^* - y_n)^2 =$$

$$= \frac{1}{2} \frac{\partial}{\partial w_{ji}} (y_j^* - y_j)^2 =$$

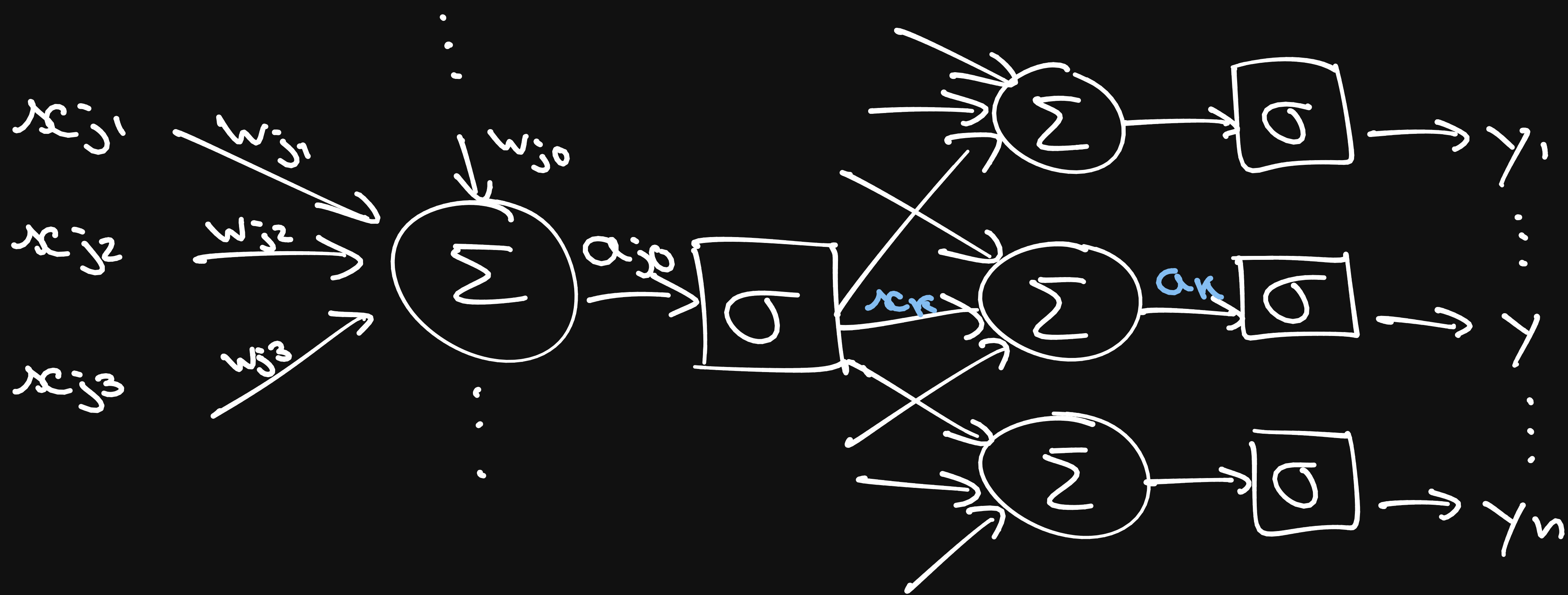
$$= \frac{1}{2} \cdot 2 \cdot (y_j^* - y_j) \left(-\frac{\partial y_j}{\partial w_{ji}} \right) =$$

$$= \left[-(\gamma_j^* - \gamma_j) \cdot \sigma'(a_j) \right] x_{ji} \\ \delta_j$$

$$= \delta_j \cdot x_{ji}$$

$$\gamma_j = \sigma\left(\sum_k w_{jk} \cdot x_{jk}\right)$$

$$\frac{\partial \gamma_j}{\partial w_{ji}} = \sigma'(a_j) \cdot x_{ji}$$



$$E = \frac{1}{2} \sum_n (\gamma_n^* - \gamma_n)^2$$

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

depende
de w_{ji}

Calcular

$$\frac{\partial E}{\partial w_{ji}} = \frac{\partial E}{\partial a_j} \cdot \frac{\partial a_j}{\partial w_{ji}}$$

$$\frac{\partial E}{\partial w_{ji}} = \frac{\partial E}{\partial a_j} \cdot \frac{\partial a_j}{\partial w_{ji}}$$

$$= \left(\sum_k \frac{\partial E}{\partial a_k} \cdot \frac{\partial a_k}{\partial a_j} \right) \frac{\partial a_j}{\partial w_{ji}}$$

$$\delta_j = - \frac{\partial E}{\partial a_j}$$

$$= -(\gamma_j^* - \gamma_j) \cdot \sigma'(a_j)$$

$$= \left(\sum_k (-\delta_k) \frac{\partial a_k}{\partial a_j} \right) \frac{\partial a_j}{\partial w_{ji}}$$

$$= \left(-\sum_k \delta_k \cdot \frac{\partial a_k}{\partial a_j} \right) \frac{\partial a_j}{\partial w_{ji}}$$

$$= \left(-\sum_k \delta_k \cdot w_{kj} \cdot \sigma'(a_j) \right) \frac{\partial a_j}{\partial w_{ji}} = w_{kj} \cdot \sigma'(a_j)$$

δ_j

$$\frac{\partial a_k}{\partial a_j} = \frac{\partial a_k}{\partial x_{kj}} \cdot \frac{\partial x_{kj}}{\partial a_j}$$

